Limits & Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Tangent lines**

The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).

$f(3) = 9 - 6 = 3$. $f'(x) = 2x - 2$, so $f'(3) = 4$. $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Evaluate $\lim_{x \to 2} (3x^2 - 4x + 1)$.

$3(4) - 8 + 1 = 12 - 8 + 1 = 5$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

$\frac{(x-3)(x+3)}{x-3} = x + 3 \to 6$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Evaluate $\lim_{x \to \infty} \frac{3x^2 + 2x}{x^2 - 5}$.

The degrees match, so the limit is the ratio of leading coefficients: 3.

5 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to 2} (3x^2 - 4x + 1)$.

$3(4) - 8 + 1 = 12 - 8 + 1 = 5$.

6 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to -1} (x^3 + 2x - 5)$.

$(-1)^3 + 2(-1) - 5 = -1 - 2 - 5 = -8$.

7 **[Foundational]** • **Target: Evaluating limits by direct substitution**

Evaluate $\lim_{x \to 0} \frac{5x + 7}{x + 3}$.

$\frac{5(0) + 7}{0 + 3} = \frac{7}{3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Evaluating limits by direct substitution

Evaluate $\lim_{x\to 3}\sqrt{x+13}$.

$\sqrt{3+13}=\sqrt{16}=4$.

9 [Intermediate] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x\to 3}\frac{x^2-9}{x-3}$.

$\frac{(x-3)(x+3)}{x-3}=x+3\to 6$.

10 [Intermediate] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x\to -2}\frac{x^2+5x+6}{x+2}$.

$\frac{(x+2)(x+3)}{x+2}=x+3\to 1$.

11 [Intermediate] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x\to 4}\frac{x^2-16}{x-4}$.

$\frac{(x-4)(x+4)}{x-4}=x+4\to 8$.

12 [Intermediate] • Target: Evaluating limits by factoring (the 0/0 form)

Evaluate $\lim_{x\to 1}\frac{x^2-1}{x^2-3x+2}$.

$\frac{(x-1)(x+1)}{(x-1)(x-2)}=\frac{x+1}{x-2}\to\frac{2}{-1}=-2$.

13 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty}\frac{3x^2+2x}{x^2-5}$.

The degrees match, so the limit is the ratio of leading coefficients: 3.

14 [Intermediate] • Target: Limits at infinity

Evaluate $\lim_{x\to\infty}\frac{2x+7}{x^2+1}$.

The denominator has higher degree, so the limit is 0.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Limits at infinity • Cross-Lesson: 2.01 Functions & Transformations

Evaluate $\lim_{x\to\infty}\frac{5x^3-2}{x^3+4x}$.

The degrees match, so the limit is the ratio of leading coefficients: 5.

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16 [Challenging] • Target: One-sided limits • Cross-Lesson: 3.01 Sequences & Series

A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x \to 2^-} f(x)$.

Using the first rule: $\lim_{x \to 2^-} f(x) = 2 + 1 = 3$.

17 [Challenging] • Target: One-sided limits • Cross-Lesson: 4.01 Trigonometry

Using the same piecewise function, find $\lim_{x \to 2^+} f(x)$.

Using the second rule: $\lim_{x \to 2^+} f(x) = 2(2) - 1 = 3$.

18 [Challenging] • Target: One-sided limits • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Using the same piecewise function, does $\lim_{x \to 2} f(x)$ exist? State its value.

Yes — both one-sided limits equal 3, so $\lim_{x \to 2} f(x) = 3$.

19 [Challenging] • Target: The power rule • Cross-Lesson: 7.01 Integration & Applications

Differentiate $f(x) = x^5$.

$f'(x) = 5x^4$.

20 [Challenging] • Target: The power rule • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

$f'(x) = 12x^3 - 4x$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.

a. $y = 3x - 9$

b. $y = 4x - 3$

c. $y = 4x + 9$

d. $y = 4x - 9$ ✓

$f(3) = 3$, $f'(3) = 4$: $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.

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22 [Foundational] • Target: MCQ — Tangent lines

The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.

a. 4 ✓

b. 2

c. 1

d. 8

The tangent line rises 4 for every 1 across.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\lim_{x \to 3} (2x^2 - 5x + 3)$.

a. 12

b. -6

c. 18

d. 6 ✓

$2(9) - 5(3) + 3 = 18 - 15 + 3 = 6$.

24 [Foundational] • Target: MCQ — Evaluating limits by direct substitution

Evaluate $\lim_{x \to 3} (2x^2 - 5x + 3)$.

a. 12

b. -6

c. 18

d. 6 ✓

$2(9) - 5(3) + 3 = 18 - 15 + 3 = 6$.

25 [Foundational] • Target: MCQ — Evaluating limits by direct substitution

Evaluate $\lim_{x \to 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$ ✓

$\frac{16 - 12}{9} = \frac{4}{9}$.

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26 [Foundational] • Target: MCQ — Limits at infinity

Evaluate $\lim_{x\to\infty} \frac{4x^2+3x}{2x^2-7}$.

a. 2 ✓

b. 0

c. $\frac{3}{7}$

d. 4

Same degree: ratio of leading coefficients $= \frac{4}{2} = 2$.

27 [Foundational] • Target: MCQ — One-sided limits

A function is defined by $f(x) = x + 2$ for $x < 3$, and $f(x) = 3x - 4$ for $x \geq 3$. Find $\lim_{x\to 3^-} f(x)$.

a. 3

b. 4

c. 9

d. 5 ✓

Using $x < 3$: $f(x) \to 3 + 2 = 5$.

28 [Foundational] • Target: MCQ — The power rule

Differentiate $f(x) = x^7$.

a. x^6

b. $7x^8$

c. $6x^7$

d. $7x^6$ ✓

Power rule: $7x^{7-1} = 7x^6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — The product rule

Differentiate $f(x) = x^3(2x + 5)$ using the product rule.

a. $8x^3 + 15x^2$ ✓

b. $6x^2 + 5$

c. $8x^3 + 5x^2$

d. $2x^3 + 15x^2$

$f'(x) = 3x^2(2x + 5) + x^3(2) = 6x^3 + 15x^2 + 2x^3 = 8x^3 + 15x^2$.

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30 [Intermediate] • Target: MCQ — The chain rule

Differentiate $f(x) = (2x + 3)^5$.

a. $5(2x + 3)^4$

b. $2(2x + 3)^4$

c. $10(2x + 3)^5$

d. $10(2x + 3)^4$ ✓

$f'(x) = 5(2x + 3)^4 \cdot 2 = 10(2x + 3)^4$.

31 [Intermediate] • Target: MCQ — The chain rule

Differentiate $f(x) = (x^3 + 2)^2$.

a. $6x^2(x^3 + 2)$ ✓

b. $6x^2(x^3 + 2)^2$

c. $3x^2(x^3 + 2)$

d. $2(x^3 + 2)$

$f'(x) = 2(x^3 + 2) \cdot 3x^2 = 6x^2(x^3 + 2)$.

32 [Intermediate] • Target: MCQ — Tangent lines

Find the slope of the tangent line to $f(x) = x^2$ at $x = 4$.

a. 4

b. 2

c. 16

d. 8 ✓

$f'(x) = 2x$; $f'(4) = 8$.

33 [Intermediate] • Target: MCQ — Increasing, decreasing, and the second derivative

For $f(x) = -x^2 + 8x$, determine the interval where f is decreasing.

a. $x < 4$

b. $x > 8$

c. $x > 4$ ✓

d. $x < 8$

$f'(x) = -2x + 8 < 0 \Rightarrow x > 4$.

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34 **[Intermediate]** • **Target: MCQ — Word problems: rates of change**

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$ ✓

b. $t = 5$

c. $t = 6$

d. $t = 2$

$h'(t) = -10t + 30 = 0 \Rightarrow t = 3$ seconds.

35 **[Intermediate]** • **Target: MCQ — One-sided limits**

Evaluate $\lim_{x \to -1} (2x^2 + 8x - 9)$.

a. -18

b. 15

c. -12

d. -15 ✓

$2(-1)^2 - 8 - 9 = -15$.

36 **[Intermediate]** • **Target: MCQ — The power rule**

Differentiate $f(x) = x^4 - 2x$.

a. $5x^3 - 2$

b. $4x^3 - 2$ ✓

c. $4x^4 - 2$

d. $x^4 - 2$

$f'(x) = 4x^3 - 2$.

37 **[Intermediate]** • **Target: MCQ — The product rule**

Differentiate $f(x) = 3\cos x$.

a. $-3\sin x$ ✓

b. $3\cos x$

c. $3\sin x$

d. $4 - \sin x$

$f'(x) = -3\sin x$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — The chain rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\lim_{x \to -1} \frac{x^2 - 2x - 3}{x + 1}$.

a. 4

b. -4 ✓

c. -6

d. -2

$\frac{(x+1)(x-3)}{x+1} = x - 3 \to -4$.

39 [Challenging] • Target: MCQ — Derivatives of trigonometric and exponential functions • Cross-Lesson: 7.01 Integration & Applications

Differentiate $f(x) = (3x + 6)(-2x - 6)$ using the product rule.

a. $-12x - 30$ ✓

b. $-12x + 30$

c. $-12x - 28$

d. $12x - 30$

$f'(x) = (3)(-2x - 6) + (3x + 6)(-2) = -6x - 18 - 6x - 12 = -12x - 30$.

40 [Challenging] • Target: MCQ — Tangent lines • Cross-Lesson: 8.01 Exponentials & Logarithms

Differentiate $f(x) = (-3x - 4)^5$.

a. $-18(-3x - 4)^4$

b. $-15(-3x - 4)^5$

c. $-15(-3x - 4)^4$ ✓

d. $5(-3x - 4)^4$

$f'(x) = 5(-3x - 4)^4(-3) = -15(-3x - 4)^4$.

41 [Challenging] • Target: MCQ — Increasing, decreasing, and the second derivative • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\lim_{x \to \infty} \frac{4x^2 + 8x}{3x^2 - 8}$.

a. 0

b. $\frac{7}{3}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$ ✓

The degrees match, so the limit is the ratio of leading coefficients: $\frac{4}{3}$.

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42 [Challenging] • Target: MCQ — Word problems: rates of change • Cross-Lesson: 1.01 Algebra & Equations

Find the equation of the tangent line to $f(x) = 2x^2 + 3x - 1$ at $x = -3$.

a. $y = 9x - 19$

b. $y = -9x + 19$

c. $y = -9x - 16$

d. $y = -9x - 19$ ✓

$f(-3) = 8$, $f'(-3) = -9$. $y - 8 = -9(x - (-3)) \Rightarrow y = -9x - 19$.

43 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Functions & Transformations

For $f(x) = -x^2 - 3$, determine the interval where f is increasing.

a. $x < 0$ ✓

b. $x > 2$

c. $x < 2$

d. $x > 0$

$f'(x) = -2x$. Increasing when $f'(x) > 0$, i.e. $x < 0$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Sequences & Series

Evaluate $\lim_{x \to 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$ ✓

$\frac{16 - 12}{9} = \frac{4}{9}$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Trigonometry

Evaluate $\lim_{x \to 5} \frac{x^2 + x - 30}{x - 5}$.

a. 11 ✓

b. 13

c. -11

d. 9

$\frac{(x - 5)(x + 6)}{x - 5} = x + 6 \to 11$.

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